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BE Computer B.

Natural Language Processing Assignment.

Q1] Write a note on WordNet -

→ • Wordnet is a large lexical database of English developed for NLP. It groups words into synsets where each synset represents a single concept. Unlike a dictionary, wordnet focuses on semantic relationships between words.

• Each word may have multiple meanings and wordnet organizes them using :-

- Synonymy - word with same meaning.
- Antonymy - opposite meaning.
- Hypernym - general term (vehicle → car).
- Hyponym - specific term (car → sedan).
- Meronymy - part of relation. (wheel → car).

* Wordnet is highly useful in -

- Wordsense disambiguation (WSD).
- Information retrieval systems.
- Machine translation.
- Text classification and semantic analysis.

- It improves understanding of context by linking words based on meaning rather than just form

Q2]

List the tools available for the development of NLP applications? Write features NLTK and TextBlob?

→

◦ Widely used tools -

- NLTK
- Gensim.
- TextBlob.
- Stanford Core NLP.
- Spacy.
- Open NLP.

◦ Features of NLTK -

NLTK is one of the most popular python libraries for NLP.

- Provides tokenization.
- Support stemming and lemmatization.
- Performs part of speech tagging.
- Includes parsing and syntax tree generation.
- Contains large datasets like corpora and wordnet.
- Useful for research and educational purposes.

◦ Features of TextBlob -

◦ TextBlob is built on top of NLTK and is easier to use -

◦ Simple and beginner friendly API.

- Performs sentiment analysis.
- Supports translation and language detection.
- Provides POS tagging and noun phrase extraction.
- Useful for quick NLP tasks and small applications.

Q3] Describe in detail the Lesk algorithm and walkers algorithm for word sense disambiguation.

→ Lesk algorithm is a classical method for word sense disambiguation. It works on the principle of overlap between dictionary definitions.

◦ Steps -

- 1) Identify ambiguous word in sentence.
- 2) Retrieve all possible meanings from dictionary.
- 3) Compare definitions of each meaning with context words.
- 4) Select the meaning with maximum overlap.

Example - He went to bank to deposit money. Overlap with financial institution is higher than river bank.

◦ Walker algorithm -

- It uses semantic relationship from lexical database like wordnet instead of just dictionary overlap.
- It calculates similarity between words using semantic distance.
- Uses relationship like hypernym, hyponym, synonym.
- Considers entire sentence context.

- Advantages - More accurate than Lesk.
 - Uses deeper semantic understanding.
- Disadvantages - Computationally expensive.

Q4] Which type of tasks are performed by the Gensim library? give an example.

→ ◦ Gensim is a powerful python library used for unsupervised topic modeling and document similarity analysis.

- Tasks performed by Gensim -
 - Topic modeling using LDA.
 - Document similarity detection.
 - word embedding.
 - Text summarization.
 - Semantic Indexing.

- Ex - If we have thousands of news articles, gensim can group them into topics such as -
 - Sports
 - Politics
 - Technology.

This helps in organizing and analyzing large text data efficiently.

- Advantages -
 - Efficient for large datasets.
 - Requires less memory.
 - Easy integration with python NLP tools.

Q5) Write a note on sentiment analysis -

→ • Sentiment analysis is the process of identifying emotional tone or opinion in text data. It is widely used in NLP to understand user feedback.

• Types of sentiments -

- 1) Positive (good, excellent).
- 2) Negative (bad, poor)
- 3) Neutral (average).

• Approaches -

1) Rule Based approach - uses predefined lexicons.

2) Machine Learning approach - uses classifiers like naive bayes, SVM.

3) Deep Learning approach - uses RNN, LSTM, Transformers.

• Applications -

- Product review analysis.
- Social media monitoring.
- Customer feedback systems.
- Brand reputation analysis.

Q6] Explain statistical machine translation with suitable diagram and example.

→ SMT is a method of translating text based on statistical model derived from bilingual corpora. It follows formula -
Translation = $\text{argmax } P(T|S)$.

where, S = source language sentence.

T = Target language sentence.

Using Bayes rule,

$$P(T|S) = P(S|T) \times P(T).$$

Components -

- 1) Language model - ensures correct grammar.
- 2) Translation model - maps source to target words.
- 3) Decoder - selects best translation.

Example -

English - I am eating Food.

Hindi - Je mange de la nourriture.

SMT chooses translation based on highest probability.

Advantages -

- No need for hand written grammar rules.
- Learns automatically from data.
- Better scalability for large datasets.

Q7] Describe various Machine Translation approaches.

→ 1] Rule based machine Translation -

RBMT is the oldest approach and is based on linguistic rules and dictionary.

Types -

- Direct MT - word to word translation.
- Transfer based MT - convert source → intermediate → target.
- Interlingua MT - uses a language independent representation.

Working -

- Analyse grammar of source language.
- Apply translation rules.
- Generate target sentence.

2] Statistical Machine Translation -

SMT uses probability and large bilingual datasets to translate text.

Working -

- learns translation pattern from data.
- Uses probability to choose best translation.

3] Neural Machine Translation -

NMT is latest and most advanced approach using deep learning.

Working -

- Uses neural network (RNN, LSTM, Transformer)
- Processes entire sentence at once.
- Uses encoder decoder architecture.

Q8] Explain natural language generation with reference architecture.

- NLG is a subfield of NLP that focuses on converting structured data into meaningful human language text.
- NLG is the process of transforming data facts or knowledge into Natural language sentences that are understandable by humans.

* Architecture -

1) Content determination -

- Selects what information should be included.
- Filters relevant data from input.

2) Text planning -

- Organizes content logically.
- Decides order information.

3) Sentence planning -

- Converts structured data into sentences.
- Includes - 1) Lexical choice.
2) Aggregation.

4) Surface realization.

- Final step.
- Applies grammar rules.
- generates fluent sentences.